

Ex. No: 6 Semaphore Implementation

C PROGRAM

```
#include <stdio.h>

#include <stdlib.h>

#include <unistd.h>

#include <semaphore.h>

#include <sys/mman.h>

#include <sys/wait.h>

int main()

{

sem_t *sem;

sem = mmap(NULL, sizeof(sem_t),

PROT_READ | PROT_WRITE,

MAP_SHARED | MAP_ANONYMOUS,

-1, 0);

sem_init(sem, 1, 1);

if(fork() == 0)

{

sem_wait(sem);

printf("Child Process Entering Critical Section\n");

sleep(3);

printf("Child Process Leaving Critical Section\n");

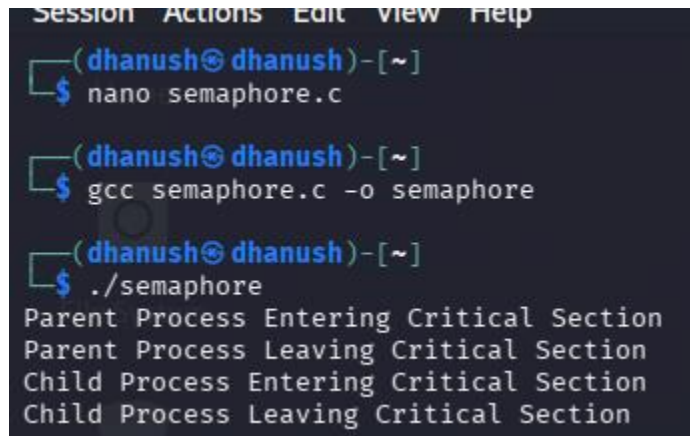
sem_post(sem);

exit(0);

}
```

```
sem_wait(sem);  
printf("Parent Process Entering Critical Section\n");  
sleep(3);  
printf("Parent Process Leaving Critical Section\n");  
sem_post(sem);  
wait(NULL);  
sem_destroy(sem);  
return 0;  
}
```

Output:

A terminal window with a dark background and light blue text. The window title is "Session Actions Edit View Help". The prompt is "(dhanush@dhanush)~". The user enters "nano semaphore.c", "gcc semaphore.c -o semaphore", and "./semaphore". The output shows the parent process entering and leaving the critical section, followed by the child process entering and leaving the critical section.

```
Session Actions Edit View Help  
(dhanush@dhanush)~  
$ nano semaphore.c  
(dhanush@dhanush)~  
$ gcc semaphore.c -o semaphore  
(dhanush@dhanush)~  
$ ./semaphore  
Parent Process Entering Critical Section  
Parent Process Leaving Critical Section  
Child Process Entering Critical Section  
Child Process Leaving Critical Section
```

Shell Script:

```
#!/bin/bash  
  
LOCKFILE="/tmp/mylock"  
  
while [ -f "$LOCKFILE" ]  
  
do  
  
sleep 1
```

done

touch "\$LOCKFILE"

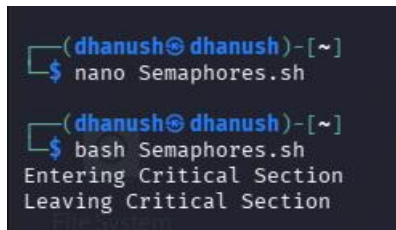
echo "Entering Critical Section"

sleep 5

echo "Leaving Critical Section"

rm -f "\$LOCKFILE"

Output:

A terminal window with a dark background and light blue text. The prompt is (dhanush@dhanush)-[~]. The user enters 'nano Semaphores.sh' and presses enter. The prompt changes to \$ bash Semaphores.sh. The script then outputs 'Entering Critical Section' and 'Leaving Critical Section' on separate lines.

```
(dhanush@dhanush)-[~]  
$ nano Semaphores.sh  
  
(dhanush@dhanush)-[~]  
$ bash Semaphores.sh  
Entering Critical Section  
Leaving Critical Section
```